

REACT: Hand Gesture Recognition from sEMG via FiLM-Conditioned User-Adaptive Contrastive Training

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<https://colab.research.google.com/drive/1chYLwW7DRUy6uFc4qvwfsmF0mOskBeVe>

Introduction

Surface electromyography (sEMG) enables gesture recognition on wearables but exhibits high inter-user variability due to anatomical differences and sensor placement, severely limiting cross generalization [1]. While CPEP [2] achieved state-of-the-art zero-shot gesture classification (78.2% in-distribution, 53.6% unseen) via contrastive pose-EMG alignment, it lacks explicit user-specific adaptation mechanisms. We extend CPEP with Feature-wise Linear Modulation (FiLM) [4] to learn user-specific embeddings from k calibration samples, enabling rapid personalization while preserving zero-shot capabilities. Deep learning is essential to model complex, non-linear mappings between noisy sEMG signals, anatomical variations, and pose representations.

Illustration

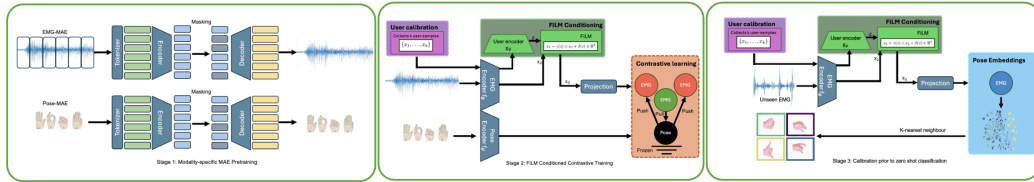


Figure 1: **FiLM-conditioned CPEP**. (1) Pre-train EMG/Pose encoders via MAE. (2) User encoder g_ψ processes k calibration embeddings to generate z ; contrastive learning jointly trains f_θ , g_ψ , and FiLM parameters (γ, β) to align FiLM-conditioned sEMG with frozen pose: $\text{FiLM}(f_\theta(x), z) = \gamma(z) \odot f_\theta(x) + \beta(z)$. (3) Inference: compute z from calibration for test samples.

Background & Related Work

sEMG Benchmarks. emg2pose [1] provides 193 users, 370 hours with supervised baselines (vemg2pose, NeuroPose, SensingDynamics) showing limited generalization. **CPEP** [2] pioneered zero-shot classification via contrastive alignment but lacks user adaptation. **Feature Modulation.** FiLM [4] conditionally modulates features via learned affine transforms, proven effective for domain adaptation [6]. **Meta-learning.** MAML [5] enables few-shot adaptation but is computationally expensive; our FiLM approach provides efficient user conditioning. **Biosignal Learning.** Recent work [3, 7] demonstrates transfer learning for biosignals, motivating our user-adaptive framework.

Data Processing

We use emg2pose [1]: 16-channel sEMG at 2kHz with synchronized pose labels (193 users, 29 gestures). Following CPEP [2], we select 4 in-distribution classes (thumb swipes, hand claw/grasp, thumb rotations, finger pinches) and 4 unseen classes (Hook/OK/Scissors, Shaka/Vulcan, Counting, Counting+wiggling). Preprocessing: (1) instance normalization per channel, (2) bandpass filter (2-250 Hz), (3) notch filter (60 Hz), (4) extract 2s windows (4000 samples). For each user, randomly sample

$k \in \{1, 2, \dots, 10\}$ windows as calibration set. Channel rotation augmentation improves placement robustness [1]. Splits: 158 training users, 15 validation, 20 test (zero overlap).

Architecture

Problem Formulation. Given paired data $\mathcal{D} = \{(x_i, p_i, y_i)\}_{i=1}^N$ where $x_i \in \mathbb{R}^{16 \times T}$ is sEMG, $p_i \in \mathbb{R}^{20 \times T}$ is pose, $y_i \in \mathcal{Y}$ is gesture label. Goal: learn representations enabling zero-shot classification on unseen gestures $\mathcal{Y}_{\text{unseen}}$ and users.

Stage 1: MAE Pre-training. We independently pre-train Transformer encoders $f_\theta^{\text{EMG}}, f_\phi^{\text{pose}}$ via masked autoencoding [8]. For each modality, we partition inputs into patches (sizes $S_{\text{EMG}} = 50$, $S_{\text{pose}} = 200$), randomly mask $r = 0.5$ of patch embeddings (mask set M), and train encoder $f_\theta^{\text{EMG}}, f_\phi^{\text{pose}}$ with decoder $\psi_\theta^{\text{EMG}}, \psi_\phi^{\text{pose}}$ to reconstruct masked patches:

$$\mathcal{L}_{\text{MAE}} = \frac{1}{|M|} \sum_{i \in M} \|\psi(f(\{z_j\}_{j \notin M}))_i - z_i\|_2^2 \quad (1)$$

where z_i denotes the i -th patch embedding. Both encoders use embedding dimension $d = 256$, 4 layers, and are trained for 100 epochs. These pre-trained weights initialize Stage 2.

Stage 2: FiLM-Conditioned Contrastive Training. Building on the pre-trained encoders, we align EMG and pose representations while adapting to individual users. We introduce a user encoder $g_\psi : \{\mathbb{R}^d\}^k \rightarrow \mathbb{R}^c$, a 3-layer Transformer with learnable query token that processes k calibration EMG embeddings $\{f_\theta(y_i)_{[\text{CLS}]}\}_{i=1}^k$ to output user embedding $z \in \mathbb{R}^c$ ($c = 128$). For each training sample (x, p) , we compute:

$$x_2 = f_\theta(x) \in \mathbb{R}^d \quad (\text{EMG encoding}) \quad (2)$$

$$x_3 = \gamma(z) \odot x_2 + \beta(z) \in \mathbb{R}^d \quad (\text{FiLM modulation}) \quad (3)$$

$$u = h(x_3) \in \mathbb{R}^d, \quad v = f_\phi(p) \in \mathbb{R}^d \quad (\text{projections}) \quad (4)$$

where $\gamma, \beta : \mathbb{R}^c \rightarrow \mathbb{R}^d$ are MLPs producing user-specific affine transformations, and h is a projection head. We freeze the pose encoder f_ϕ and optimize symmetric InfoNCE [9]:

$$\mathcal{L}_{\text{CPEP}} = -\frac{1}{2N} \sum_{i=1}^N \left[\log \frac{\exp(s_{ii})}{\sum_j \exp(s_{ij})} + \log \frac{\exp(s_{ii})}{\sum_j \exp(s_{ji})} \right] \quad (5)$$

where $s_{ij} = \frac{\tilde{u}_i^\top \tilde{v}_j}{\tau}$, $\tilde{u} = u/\|u\|$, and $\tau = 0.02$ is learnable temperature. At each iteration, for each user: sample k calibration EMGs to compute z , apply FiLM conditioning to that user's samples, and backpropagate $\mathcal{L}_{\text{CPEP}}$ to update $f_\theta, g_\psi, \gamma, \beta$, and h (batch 256, AdamW lr= 10^{-4} , wd= 10^{-5} , 100 epochs).

Stage 3: Inference. At test time, given calibration set $\{y_1, \dots, y_k\}$, we compute user embedding $z = g_\psi(\{f_\theta(y_i)_{[\text{CLS}]}\}_{i=1}^k)$ once. For each test sample x_{test} , we compute $u_{\text{test}} = h(\text{FiLM}(f_\theta(x_{\text{test}}), z))$ and perform zero-shot classification by retrieving the top-10 nearest pose embeddings via cosine similarity, predicting via majority vote.

Baseline Model

Baselines: (1) CPEP [2] without user conditioning, (2) vemg2pose, NeuroPose, SensingDynamics [1] via linear probing, (3) EMG-MAE (self-supervised only), (4) PoseT (supervised pose regression). **Metrics:** Macro accuracy on in-distribution (4 classes) and unseen (4 classes) gestures. Evaluate via linear probing (LP) and zero-shot (ZS, $k = 10$ -NN).

Ethical Considerations

Privacy: emg2pose collected under IRB with consent [1]; all data de-identified. sEMG may encode health information but we limit the analysis to gesture recognition. **Coded gaze:** Performance disparities may exist across demographics due to data misrepresentation [10]. **Limitations:** Dataset includes primarily healthy adults. Generalization to elderly or neuromuscular conditions is unexplored. Future should expand to diverse populations and underrepresented groups.

References

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